

B.E. Information Technology — Distributed Systems

(414450)

Question Frequency Analysis · All 5 Papers · 2019 Pattern · Sem VIII

Paper	Exam Period	Paper Code
P1	MAR–APR 2023	[6004]-574
P2	MAR–APR 2024	[6263]-181
P3	MAR–APR 2025	[6404]-174
P4	NOV–DEC 2023	[6181]-202
P5	NOV–DEC 2025	[6584]-163

Color Key: **Red freq (3x+)** = Highest priority | **Blue freq (2x)** = Moderate priority | **Yellow rows** = Appeared 4–5 times (must-prepare) | **Light rows** = Appeared 1–3 times

Unit 1 <i>Q1 & Q2 — Communication, Clocks & Synchronization</i>			
#	Topic / Question	Appeared In	Freq.
1	XDR, Marshalling & Unmarshalling + 3 Alternative Approaches	Papers 2, 3, 5	3x
2	Lamport's Logical Clock (Algorithm, Timestamps, Happened-Before)	Papers 1, 2, 5	3x
3	Clock Synchronization — Berkeley Algorithm + NTP	Papers 3, 4	2x
4	IPC Characteristics (Sync/Async, Blocking Receive in Java)	Papers 1, 5	2x
5	Overlay Networks (Types, Advantages, Skype)	Papers 1, 2	2x
6	Election Algorithm & Bully Algorithm	Papers 1, 3	2x
7	Message Passing Interface (MPI) — Architecture, Point-to-Point, Send Ops	Papers 3, 4	2x
8	Happened-Before Relationship & Lamport Timestamp Numerical	Papers 2, 5	2x
9	Mutual Exclusion — Permission-based Centralized Algorithm	Paper 2	1x

Unit 2 <i>Q3 & Q4 — Replication, Consistency & Fault Tolerance</i>			
#	Topic / Question	Appeared In	Freq.
1	Coordinated Checkpointing Recovery Mechanism	Papers 1, 2, 3, 4	4x
2	Causal Consistency Model with Distributed Shared Database	Papers 1, 2, 3	3x
3	Reasons for Replication & Relation to Scalability	Papers 2, 3, 5	3x
4	Replica Management — Permanent, Server-Initiated, Client-Initiated Replicas	Papers 1, 4	2x
5	Primary-Based Protocol & Replicated Write Protocols (Active Replication)	Papers 1, 3	2x
6	Two-Phase Commit Protocol & Distribution Commit Problem	Papers 4, 5	2x
7	Fault Tolerance — Fault Classes (Transient, Intermittent, Permanent) & Failure Types	Paper 2	1x
8	Dependable Systems & RPC Communication Failure Handling	Paper 4	1x
9	Client-Centric Consistency — Monotonic Read Consistency	Paper 5	1x

#	Topic / Question	Appeared In	Freq.
10	Replica Server Placement & Content Placement Strategies	Paper 5	1×
11	Primary-Backup Protocol	Paper 4	1×

Unit 3 <i>Q5 & Q6 — Distributed File Systems, Naming & Multimedia</i>			
#	Topic / Question	Appeared In	Freq.
1	Sun Network File System (NFS) Architecture	Papers 1, 3, 5	3×
2	Directory Service — DNS vs X.500, X.500 Service Architecture	Papers 1, 2, 3	3×
3	Web Services & Apache Web Server Organization	Papers 1, 2, 3	3×
4	P2P Download System & BitTorrent Protocol	Papers 1, 3, 5	3×
5	QoS Management in Distributed Multimedia Systems	Papers 4, 5	2×
6	Web Service in Horizontal Distribution using Web Server Clusters	Papers 4, 5	2×
7	Hadoop Distributed File System (HDFS) — Master-Slave Architecture	Paper 2	1×
8	Multimedia Streaming — Bandwidth, Latency, Loss Rate & QoS Negotiation	Paper 2	1×
9	Distributed File System — Key Design Issues & Requirements	Paper 4	1×
10	Client-Side Web Caching & Cooperative Caching	Paper 4	1×

Unit 4 <i>Q7 & Q8 — SOA, Monitoring Tools, Wearables & Microkernels</i>			
#	Topic / Question	Appeared In	Freq.
1	Mach & CHORUS Microkernels	Papers 1, 2, 3, 4, 5	5×
2	Service Oriented Architecture (SOA) — Components & Application	Papers 1, 2, 4, 5	4×
3	Wearable Devices, PVM & JINI (Brief Explanations)	Papers 1, 3, 5	3×
4	Zabbix — Key Features / Overview	Papers 1, 2, 4	3×
5	Nagios — Key Features / Overview	Papers 1, 2, 3	3×
6	Prometheus — Data Model, Query Language, Alerting Rules	Papers 3, 4, 5	3×
7	Wearable Devices in Distributed Systems & Wearable Computing Problems	Paper 2	1×
8	Mach & CHORUS — Memory Management (Avoid Physical Copying)	Paper 4	1×

★ Yellow rows = appeared in 4–5 papers (highest priority for exam preparation) | Frequency counted across 5 SPPU question papers (2023–2025)